CONICS AND $(q+1)/2$ EXTERIOR POINTS FORMING A SET WITHOUT TANGENTS: A PROOF OF A CONJECTURE OF VAN DE VOORDE

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Abstract. Let $\mathcal{C}$ be a non-degenerate conic in $\mathrm{PG}(2,q)$, q odd. Van de Voorde (2011) conjectured that the union of $\mathcal{C}$ with $(q+1)/2$ points exterior to $\mathcal{C}$ can be a set without tangents (a set met by no line in exactly one point) only for $q=7$ and $q=11$. We prove this conjecture. Equivalently, apart from the trivial case $q=3$ and the cases $q=7,11$, every exterior set of $(q+1)/2$ points of a conic contains three collinear points. For $q\equiv 1 \pmod 4$ this follows from a theorem of Blokhuis, Seress and Wilbrink; the case $q\equiv 3 \pmod 4$ is new. The proof combines an analogue of Segre's lemma of tangents, which converts the absence of collinear triples into a multiplicative identity for the quadratic character, with autocorrelation identities of the Paley matrix, Jacobi sums and Stickelberger's theorem, and Weil's bound. We also show that for $q=7$ and $q=11$ the configuration is unique up to projectivity, with stabilisers S_4 and A_5. The proofs have been checked numerically, and the main chain of the argument has been formalised in Lean 4 modulo the classical theorems it uses.

1. Introduction

Throughout, $q=p^k$ is a power of an odd prime p and $\mathrm{PG}(2,q)$ is the Desarguesian projective plane of order q. A set X of points of $\mathrm{PG}(2,q)$ is a *set without tangents* (or an *untouchable set*) if no line meets X in exactly one point. For even q hyperovals are the smallest non-empty examples; for odd q the smallest size u_q of a non-empty set without tangents is known only for a few values of q. Blokhuis, Seress and Wilbrink [2] showed $u_3=6$, $u_5=10$, $u_7=12$, and Van de Voorde [16, §1.2] found $u_9=15$ and $u_{11}=18$ by computer; see also Héger and Nagy [7] and Dover [6] for recent work on untouchable sets and their generalisations.

Let $\mathcal{C}$ be a non-degenerate conic. A point not on $\mathcal{C}$ is *exterior* if it lies on two tangent lines of $\mathcal{C}$, and a line is *external* (a *passant*) if it misses $\mathcal{C}$. An *exterior set* of $\mathcal{C}$ is a set E of exterior points such that the line joining any two points of E is external [16, §1.1]. An exterior set has at most $(q+1)/2$ points (Lemma 3.3); it is *complete* if it has exactly $(q+1)/2$ points. The exterior points on a passant form a complete exterior set; we call such sets *linear*.

Theorem 1.1 (Blokhuis, Seress, Wilbrink [3])**.** *If $q\equiv 1 \pmod 4$, every complete exterior set of a conic in $\mathrm{PG}(2,q)$ is linear.*

For $q\equiv 3 \pmod 4$ there are non-linear complete exterior sets for $q=7,11,\dots,31$, and Blokhuis, Seress and Wilbrink conjectured that every complete exterior set is linear when $q>31$ [3]; see [16, §1.1, §3]. This conjecture is still open.

For $q=7$ and $q=11$ some complete exterior sets S contain no three collinear points, and then $\mathcal{C}\cup S$ is a set without tangents of size $12=u_7$, respectively $18=u_{11}$ [16, §3], [7]. According to [16, §3], the computations of Blokhuis, Seress and Wilbrink show that for $11<q\le 31$ every complete exterior set contains three collinear points. On this basis Van de Voorde conjectured that $q=7$ and $q=11$ are the only values of q for which a conic and $(q+1)/2$ of its exterior points form a set without tangents [16, §3]. The conjecture is a

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consequence of the Blokhuis–Seress–Wilbrink conjecture: a linear complete exterior set has $(q+1)/2 \geq 3$ collinear points when $q \geq 5$. We prove it unconditionally.

Theorem A. *Let q be an odd prime power, $\mathcal{C}$ a non-degenerate conic in $\mathrm{PG}(2,q)$, and S a set of $(q+1)/2$ points exterior to $\mathcal{C}$.*

(a) *$\mathcal{C} \cup S$ is a set without tangents if and only if S is an exterior set of $\mathcal{C}$ and no three points of S are collinear.*
(b) *Such sets S exist if and only if $q \in \{3, 7, 11\}$.*
(c) *For $q \in \{7, 11\}$ the set S is unique up to collineations fixing $\mathcal{C}$: for a given conic there are exactly 14 (for $q = 7$), respectively 22 (for $q = 11$), such sets, and the stabiliser of each of them in the group $\mathrm{PGL}(2,q)$ of collineations fixing $\mathcal{C}$ is isomorphic to S_4, respectively A_5.*

Corollary B (Van de Voorde's conjecture)**.** *For odd $q \notin \{3, 7, 11\}$ a conic of $\mathrm{PG}(2,q)$ together with $(q+1)/2$ of its exterior points never forms a set without tangents. Equivalently, every complete exterior set of a conic in $\mathrm{PG}(2,q)$, $q \notin \{3, 7, 11\}$, contains three collinear points.*

The case $q = 3$ is trivial (S consists of the two exterior points of a passant) and is implicitly excluded in [16]. For $q \equiv 1 \pmod 4$, Theorem A(b) is immediate from Theorem 1.1; the content of this paper is the case $q \equiv 3 \pmod 4$, which we prove both for primes and for prime powers.

Outline of the proof. We identify exterior points of $\mathcal{C}$ with unordered pairs of points of $\mathrm{PG}(1,q)$ (the two points of tangency), using binary quadratic forms (Section 3). After a projectivity, a complete exterior set containing the point $\{0, \infty\}$ is described by a fixed-point-free involution ψ of $\mathbb{F}_q^*$ satisfying three conditions (H1)–(H3) on the quadratic character χ. Our argument has four steps.

(1) A Segre-type lemma (Lemma 4.2), in the spirit of Segre's lemma of tangents [13]: for three pairwise compatible exterior points, a product of "slopes" over the external lines through them equals a sign $-\varepsilon$ coming from the Paley tournament. If no three points of S are collinear, Ceva's theorem forces $\varepsilon = -1$, which in coordinates is the identity (A): $\chi(\psi z - \psi y) = -\chi(z-y)\chi(yz)$.
(2) Two autocorrelation identities (Lemma 5.1) give the matrix identity $\mathsf{NA} + \mathsf{AN} = -2I$ for the Jacobsthal matrix A and a matrix N supported on the graph of ψ. Evaluated on multiplicative characters it becomes $(J(\chi,\eta) + \overline{J(\chi,\eta')})M(\eta',\eta) = -2(q-1)\delta_{\eta\eta'}$ (Lemma 5.2), where M is the matrix of ψ in the character basis.
(3) Stickelberger's theorem on the prime factorisation of Gauss sums, together with a combinatorial lemma on base-p digits (Lemma 6.1), shows that $J(\chi,\eta_1) + \overline{J(\chi,\eta')}$ vanishes, for a generator η_1, only when $\eta' = \chi\eta_1^{p^j}$ (Proposition 6.2).
(4) Hence $\psi(x) = cx^{p^j}$ on the squares (Proposition 7.2). Weil's bound then leaves only $q \in \{7, 11\}$ when $j = 0$ (Proposition 7.3) and rules out $j \neq 0$ altogether (Proposition 7.4).

Section 8 assembles the proof of Theorem A and treats the examples for $q = 7, 11$. Section 9 describes the numerical checks and a Lean 4 formalisation of the main chain of the argument.

2. Preliminaries

2.1. **Notation.** Let $\mathbb{F}_q$ be the field with $q = p^k$ elements, p odd, and χ its quadratic character, extended by $\chi(0) = 0$. We write $h = (q-1)/2$, $\mathcal{S} \subset \mathbb{F}_q^*$ for the set of non-zero squares and $\mathcal{N} = \mathbb{F}_q^* \setminus \mathcal{S}$ for the non-squares; $|\mathcal{S}| = |\mathcal{N}| = h$. We put $\mathrm{PG}(1,q) = \mathbb{F}_q \cup \{\infty\}$. For an integer $m \geq 1$ and $x \in \mathbb{Z}/m\mathbb{Z}$ (or $x \in \mathbb{Z}$) we denote by $\langle x \rangle_m \in \{0, \ldots, m-1\}$ the least non-negative residue. When $q \equiv 3 \pmod 4$ we have $p \equiv 3 \pmod 4$, k odd, h odd and $\chi(-1) = -1$.

2.2. **Character sums.** We use two standard facts.

(W0) If $a \neq 0$ and $D = b^2 - 4ac \neq 0$, then $\sum_{x\in\mathbb{F}_q} \chi(ax^2+bx+c) = -\chi(a)$. Indeed, $\chi(4a)\chi(ax^2+bx+c) = \chi((2ax+b)^2 - D)$, and $\sum_{y\in\mathbb{F}_q}\chi(y^2-D) = -1$ because $y^2 - z^2 = D$ has $q-1$ solutions.

(W) (Weil's bound) If $f \in \mathbb{F}_q[x]$ has degree $d \geq 1$ and is not of the form $a\,g(x)^2$ with $a \in \mathbb{F}_q$, $g \in \mathbb{F}_q[x]$, then $\left|\sum_{x\in\mathbb{F}_q}\chi(f(x))\right| \leq (d-1)\sqrt{q}$; see [10, Theorem 5.41], [12]. (In [10] the factor is $d'-1$, where $d' \leq d$ is the number of distinct roots of f.)

2.3. **Gauss and Jacobi sums.** Let $\widehat{\mathbb{F}_q^*}$ be the group of multiplicative characters $\eta\colon \mathbb{F}_q^* \to \mathbb{C}^*$, extended by $\eta(0) = 0$; write 1 for the trivial character. Let $\zeta_p = e^{2\pi i/p}$ and $\mathrm{Tr}\colon \mathbb{F}_q \to \mathbb{F}_p$ the absolute trace. For $\lambda, \mu \in \widehat{\mathbb{F}_q^*}$ put

$$G(\lambda) = \sum_{x\in\mathbb{F}_q} \lambda(x)\,\zeta_p^{\mathrm{Tr}(x)}, \qquad J(\lambda,\mu) = \sum_{t\in\mathbb{F}_q} \lambda(1-t)\,\mu(t).$$

We use the following standard facts (see e.g. [10, Chapter 5], [1]): if $\lambda \neq 1$ then $|G(\lambda)|^2 = q$; if $\lambda, \mu, \lambda\mu \neq 1$ then $J(\lambda,\mu) = G(\lambda)G(\mu)/G(\lambda\mu)$, so $|J(\lambda,\mu)| = \sqrt{q}$. Since χ is real, $\overline{J(\chi,\eta)} = J(\chi,\overline{\eta})$.

Lemma 2.1. *$J(\chi,\eta^p) = J(\chi,\eta)$ for every $\eta \in \widehat{\mathbb{F}_q^*}$.*

Proof. The map $t \mapsto t^p$ is a bijection of $\mathbb{F}_q$ fixing 1, and $\chi(z^p) = \chi(z)^p = \chi(z)$. Hence $J(\chi,\eta^p) = \sum_t \chi(1-t)\eta(t^p) = \sum_t \chi(1-t^p)\eta(t^p) = \sum_u \chi(1-u)\eta(u)$. □

Stickelberger's theorem. Let $K = \mathbb{Q}(\zeta_{q-1},\zeta_p)$, let $\mathfrak{p}$ be a prime of $\mathbb{Z}[\zeta_{q-1}]$ above p, whose residue field we identify with $\mathbb{F}_q$, and let $\mathfrak{P}$ be the prime of K above $\mathfrak{p}$. The *Teichmüller character* $\omega \in \widehat{\mathbb{F}_q^*}$ is defined by $\omega(x) \equiv x \pmod{\mathfrak{p}}$; it generates $\widehat{\mathbb{F}_q^*}$. For $j \in \mathbb{Z}/(q-1)\mathbb{Z}$ let $s(j)$ denote the sum of the base-p digits of $\langle j\rangle_{q-1}$.

Theorem 2.2 (Stickelberger [14]; see [17, Chapter 6])**.** *For $j \in \mathbb{Z}/(q-1)\mathbb{Z}$, $j \neq 0$, the $\mathfrak{P}$-adic valuation of $G(\omega^{-j})$, normalised by $v_{\mathfrak{P}}(p) = 1$, is $v_{\mathfrak{P}}(G(\omega^{-j})) = s(j)/(p-1)$.*

Remark 2.3. Some authors state Theorem 2.2 for ω^j instead of ω^{-j}, or define Gauss sums with an extra sign. Neither change affects our use of the theorem: valuations ignore signs, and we only use the theorem in the form of identities quantified over all $t \in (\mathbb{Z}/(q-1)\mathbb{Z})^*$, which are invariant under $t \mapsto -t$.

For $t \in (\mathbb{Z}/(q-1)\mathbb{Z})^*$ let $\sigma_t \in \mathrm{Gal}(K/\mathbb{Q}(\zeta_p))$ be the automorphism with $\sigma_t(\zeta_{q-1}) = \zeta_{q-1}^t$; it exists because $\gcd(p, q-1) = 1$, so that $\mathbb{Q}(\zeta_{q-1})$ and $\mathbb{Q}(\zeta_p)$ are linearly disjoint. Since character values are $(q-1)$-st roots of unity and ζ_p is fixed, $\sigma_t(G(\lambda)) = G(\lambda^t)$.

3. Exterior points as pairs

3.1. **The model.** Let V be the 3-dimensional $\mathbb{F}_q$-space of binary quadratic forms $f(X,Y) = aX^2 + bXY + cY^2$; we identify $\mathrm{PG}(2,q)$ with the projective space of V and write $[f]$ for the point defined by $f \neq 0$. The discriminant $\Delta(f) = b^2 - 4ac$ is a non-degenerate quadratic form on V, with polar form

$$B(f,g) = b_f b_g - 2(a_f c_g + c_f a_g), \qquad \Delta(sf+ug) = s^2\Delta(f) + 2su\,B(f,g) + u^2\Delta(g).$$

Hence $\mathcal{C} = \{[f] : \Delta(f) = 0\}$ is a non-degenerate conic; since all non-degenerate conics of $\mathrm{PG}(2,q)$ are projectively equivalent [8], we may and do work with this one. Its points are the squares $[\ell_t^2]$ of linear forms, where $\ell_t = X - tY$ for $t \in \mathbb{F}_q$ and $\ell_\infty = Y$. For $f \in V$ we write $f(t) = f(t,1)$ for $t \in \mathbb{F}_q$ and $f(\infty) = f(1,0) = a_f$; thus $f(t) = 0$ if and only if $\ell_t \mid f$.

The group $\mathrm{GL}(2,q)$ acts on V by linear substitutions, and $\Delta(f\circ A) = \det(A)^2\Delta(f)$. This gives an action of $\mathrm{PGL}(2,q)$ on $\mathrm{PG}(2,q)$ by collineations fixing $\mathcal{C}$, compatible with the action of $\mathrm{PGL}(2,q)$ on $\mathrm{PG}(1,q)$ by Möbius transformations on the roots of the forms. For prime q this is the full stabiliser of $\mathcal{C}$ in the collineation group [8].

Lemma 3.1. (i) *The tangent line to $\mathcal{C}$ at $[\ell_t^2]$ is $T_t = \{[f] : f(t) = 0\}$.*

(ii) *A point* $[f] \notin \mathcal{C}$ *is exterior if and only if* $f = \kappa\,\ell_a\ell_b$ *with* $a \neq b$ *in* $\mathrm{PG}(1,q)$ *and* $\kappa \in \mathbb{F}_q^*$*; the tangents through it are* T_a *and* T_b*. Thus exterior points correspond bijectively to* 2*-subsets* $\{a,b\}$ *of* $\mathrm{PG}(1,q)$*; we write* $P = \{a,b\}$ *and* $F_P = \ell_a\ell_b$*.*

(iii) *Let* $P = \{p_1,p_2\}$ *be an exterior point and* $f \in V$ *with* $[f] \neq P$*. Then*

$$B(F_P,f)^2 - \Delta(F_P)\Delta(f) = 4f(p_1)f(p_2), \tag{3.1}$$

and the line $P[f]$ *is a tangent if* $f(p_1)f(p_2) = 0$*, a secant if* $f(p_1)f(p_2) \in \mathcal{S}$*, and a passant if* $f(p_1)f(p_2) \in \mathcal{N}$*.*

(iv) *Through an exterior point there pass* 2 *tangents,* h *secants and* h *passants of* $\mathcal{C}$*.*

(v) *Two distinct exterior points* P,Q *sharing an element of* $\mathrm{PG}(1,q)$ *are joined by a tangent. If* $P = \{a,b\}$ *and* $Q = \{c,d\}$ *are disjoint, the line* PQ *is a passant if and only if* $\chi(R(P,Q)) = -1$*, where* $R(P,Q) = (a-c)(a-d)(b-c)(b-d)$ *and factors involving* ∞ *are omitted.*

Proof. (i) T_t is a line containing $[\ell_t^2]$, and $[\ell_s^2] \in T_t$ only for $s = t$; so it is the tangent at $[\ell_t^2]$.

(ii) By (i), $[f]$ lies on a tangent if and only if f has a root in $\mathrm{PG}(1,q)$, i.e. splits into linear factors over $\mathbb{F}_q$. Since $[f] \notin \mathcal{C}$, the two factors are distinct.

(iii) Identity (3.1) is a polynomial identity in p_1, p_2 and the coefficients of f; it is checked by direct expansion (separately for $p_2 = \infty$, where $F_P = \ell_{p_1}Y$). The line $P[f]$ meets $\mathcal{C}$ in the points $[sF_P + uf]$ with $\Delta(sF_P + uf) = 0$. The binary quadratic form $(s,u) \mapsto \Delta(sF_P + uf)$ is not identically zero (a line is not contained in $\mathcal{C}$), and its discriminant is $4(B(F_P,f)^2 - \Delta(F_P)\Delta(f)) = 16f(p_1)f(p_2)$. So it has 1, 2 or 0 zeros in $\mathrm{PG}(1,q)$ according as $f(p_1)f(p_2)$ is zero, a non-zero square or a non-square.

(iv) Choose g_1, g_2 with F_P, g_1, g_2 a basis of V. The lines through P are the lines $P[yg_1 + zg_2]$, $(y:z) \in \mathrm{PG}(1,q)$, and

$$n_P(y:z) = \frac{y\,g_1(p_1) + z\,g_2(p_1)}{y\,g_1(p_2) + z\,g_2(p_2)} \in \mathrm{PG}(1,q)$$

is a Möbius function of $(y:z)$. It is invertible: otherwise a non-zero combination $yg_1 + zg_2$ would vanish at p_1 and p_2, hence be a multiple of F_P. By (iii), the line is a tangent, a secant or a passant according as $n_P \in \{0,\infty\}$, $n_P \in \mathcal{S}$ or $n_P \in \mathcal{N}$, which proves the count.

(v) If P,Q share a, both lie on T_a. Otherwise apply (iii) with $f = F_Q$: $F_Q(a)F_Q(b) = R(P,Q)$. □

We call disjoint pairs P,Q with $\chi(R(P,Q)) = -1$ *compatible*. Since $R(P,Q)$ has an even number of factors, compatibility is symmetric.

3.2. **Sets without tangents and exterior arcs.** A set of points with no three collinear is an *arc*.

Proposition 3.2 (Theorem A(a))**.** *Let* S *be a set of* $(q+1)/2$ *exterior points of* $\mathcal{C}$*. Then* $\mathcal{C} \cup S$ *is a set without tangents if and only if* S *is an exterior set and an arc.*

Proof. Suppose $\mathcal{C} \cup S$ has no tangents. Each of the $q+1$ tangent lines of $\mathcal{C}$ meets $\mathcal{C}$ in one point, so it contains a point of S. Each point of S lies on exactly two tangents, so there are $2 \cdot \frac{q+1}{2} = q+1$ incidences, and every tangent contains exactly one point of S. Let $P \in S$. Each of the h passants through P (Lemma 3.1(iv)) must contain a point of $S \setminus \{P\}$, a set of h points. Hence each of these passants contains exactly one point of $S \setminus \{P\}$, and all points of $S \setminus \{P\}$ lie on passants through P. So all lines PQ, $Q \in S$, are passants, and no line through P contains two further points of S.

Conversely, let S be an exterior set and an arc. A tangent contains at most one point of S, since the line joining two points of S is a passant; by the count above it contains exactly one, so it meets $\mathcal{C} \cup S$ in two points. A secant meets $\mathcal{C}$ twice. If a passant contains $P \in S$, then, since S is an arc and an exterior set, the h lines PQ, $Q \in S \setminus \{P\}$, are h distinct passants through P, i.e. all of them; so the passant contains exactly one further point of S. A passant missing S misses $\mathcal{C} \cup S$. □

Lemma 3.3. *The points of an exterior set E are pairwise disjoint and pairwise compatible 2-subsets of* $\mathrm{PG}(1,q)$. *In particular* $|E| \le (q+1)/2$, *with equality if and only if E is the set of 2-cycles of a fixed-point-free involution of* $\mathrm{PG}(1,q)$.

Proof. This is Lemma 3.1(v). □

3.3. **Normal form.** Let S be a complete exterior set. Since $\mathrm{PGL}(2,q)$ is transitive on 2-subsets of $\mathrm{PG}(1,q)$, we may assume that $P_0 = \{0,\infty\} \in S$. By Lemma 3.3,

$$S = \{P_0\} \cup \{\{x, \psi(x)\} : x \in \mathbb{F}_q^*\}$$

for a map $\psi\colon \mathbb{F}_q^* \to \mathbb{F}_q^*$, and by Lemma 3.1(v) the following hold:

(H1) ψ is an involution without fixed points;
(H2) $\chi(\psi(x)) = -\chi(x)$ for all $x \in \mathbb{F}_q^*$ (compatibility with P_0, as $R(P_0, \{x, \psi x\}) = x\psi(x)$);
(H3) $\chi((x-y)(x-\psi y)(\psi x - y)(\psi x - \psi y)) = -1$ for all $x, y \in \mathbb{F}_q^*$ with $y \notin \{x, \psi x\}$.

Conversely, every ψ with (H1)–(H3) defines a complete exterior set containing P_0. For example, $\psi(x) = n/x$ with $n \in \mathcal{N}$ satisfies (H1)–(H3), and the corresponding set is linear. In this language Theorem 1.1 states that for $q \equiv 1 \pmod 4$ every ψ with (H1)–(H3) has this form.

4. A Segre-type lemma

From now on $q \equiv 3 \pmod 4$, so $\chi(-1) = -1$ and h is odd.

4.1. **The Paley orientation.** For distinct $x, y \in \mathrm{PG}(1,q)$ put $\theta(x,y) = \chi(x-y)$ if $x, y \in \mathbb{F}_q$, $\theta(\infty, y) = 1$, $\theta(x, \infty) = -1$, and for pairwise distinct x, y, z put $O(x,y,z) = \theta(x,y)\theta(y,z)\theta(z,x)$. Since $\chi(-1) = -1$, θ and O are antisymmetric.

Lemma 4.1. (i) *Let $g \in \mathrm{PGL}(2,q)$ be represented by a matrix with determinant* $\det g$. *Then $O(gx, gy, gz) = \chi(\det g)\, O(x,y,z)$.*
(ii) *For pairwise compatible exterior points $P = \{a, a'\}$, $Q = \{b, b'\}$, $R = \{c, c'\}$ the sign $\varepsilon(P,Q,R) = O(a,b,c)\, O(a',b',c')$ does not depend on the labelling of the elements inside the pairs, and is invariant under* $\mathrm{PGL}(2,q)$.
(iii) *In the frame of Section 3, $\varepsilon(P_0, \{y, \psi y\}, \{z, \psi z\}) = \chi((\psi z - \psi y)(z-y))\chi(yz)$.*

Proof. (i) It suffices to check the generators $x \mapsto x + b$ (determinant 1; O is invariant), $x \mapsto ax$ (determinant a; each factor θ between finite points is multiplied by $\chi(a)$, and $O(\infty, y, z) = -\chi(y-z)$ also changes by $\chi(a)$) and $x \mapsto -1/x$ (determinant 1). For the last one: for non-zero finite x, y, z, $\chi(-1/x + 1/y) = \chi(x-y)\chi(xy)$, and the factors $\chi(xy)\chi(yz)\chi(zx)$ form a square; moreover $O(0,y,z) = -\chi(yz)\chi(y-z) = O(\infty, -1/y, -1/z)$, $O(\infty, y, z) = -\chi(y-z) = O(0, -1/y, -1/z)$ and $O(0, \infty, z) = -\chi(z) = O(\infty, 0, -1/z)$.

(ii) Swapping $a \leftrightarrow a'$ multiplies ε by

$$\theta(a,b)\theta(a,b')\theta(a',b)\theta(a',b') \cdot \theta(c,a)\theta(c',a)\theta(c,a')\theta(c',a') = \chi(R(P,Q))\,\chi(R(P,R)) = 1$$

(if ∞ occurs, the omitted factors of R correspond to values $\theta = \pm 1$ occurring in pairs). Invariance follows from (i), since both factors of ε are multiplied by $\chi(\det g)$.

(iii) $O(0,y,z) = -\chi(yz)\chi(y-z)$ and $O(\infty, \psi y, \psi z) = -\chi(\psi y - \psi z)$. □

4.2. **The lemma.** Let P, Q, R be pairwise compatible, non-collinear exterior points, and fix representatives F_P, F_Q, F_R; they form a basis of V. Every line through P other than PQ and PR meets QR in a point $[yF_Q + zF_R]$ with $yz \neq 0$; put $\lambda_P = z/y$. Similarly, for lines through Q meeting PR in $[xF_P + zF_R]$ put $\mu_Q = x/z$, and for lines through R meeting PQ in $[xF_P + yF_Q]$ put $\nu_R = y/x$. Let Π_P be the product of λ_P over the passants through P other than PQ, PR, and define Π_Q, Π_R in the same way. If $[f] = [xF_P + yF_Q + zF_R]$ with $xyz \neq 0$, then the lines $P[f]$, $Q[f]$, $R[f]$ have

$$\lambda_P\, \mu_Q\, \nu_R = \frac{z}{y} \cdot \frac{x}{z} \cdot \frac{y}{x} = 1. \tag{4.1}$$

Lemma 4.2 (Segre-type lemma)**.** $\Pi_P\, \Pi_Q\, \Pi_R = -\varepsilon(P,Q,R)$.

Proof. Both sides are invariant under $\mathrm{PGL}(2,q)$: the action is linear on V, so it maps representatives to representatives, preserves the coordinates x,y,z and preserves passants; for the right-hand side use Lemma 4.1(ii). As $q+1>6$, we may therefore assume that ∞ is not among the six elements p_1,p_2,q_1,q_2,r_1,r_2 of $P=\{p_1,p_2\}$, $Q=\{q_1,q_2\}$, $R=\{r_1,r_2\}$. Each representative is $F_W=\kappa_W\ell_{w_1}\ell_{w_2}$ with $\kappa_W\in\mathbb{F}_q^*$.

For a line through P meeting QR in $[yF_Q+zF_R]$ put $n=n_P=(F_Q(p_1)+\lambda F_R(p_1))/(F_Q(p_2)+\lambda F_R(p_2))$, $\lambda=z/y$, as in the proof of Lemma 3.1(iv). The line PQ has $n=n_Q:=F_Q(p_1)/F_Q(p_2)$ and PR has $n=n_R:=F_R(p_1)/F_R(p_2)$; both are non-squares, because PQ and PR are passants (in particular $F_Q(p_i),F_R(p_i)\neq 0$). Solving for λ,

$$\lambda=c_P\,\frac{n-n_Q}{n-n_R},\qquad c_P=-\frac{F_Q(p_2)}{F_R(p_2)}.$$

By Lemma 3.1(iii),(iv), the passants through P other than PQ,PR correspond to $n\in\mathcal{N}\setminus\{n_Q,n_R\}$, a set of $h-2$ elements. Since $\prod_{n\in\mathcal{N}}(X-n)=X^h+1$, differentiating at $\alpha\neq\beta$ in $\mathcal{N}$ gives

$$\prod_{n\in\mathcal{N}\setminus\{\alpha,\beta\}}\frac{n-\alpha}{n-\beta}=-\frac{h\alpha^{h-1}}{h\beta^{h-1}}=-\frac{\beta}{\alpha}\qquad(\text{as }\alpha^h=\beta^h=-1).$$

As $h-2$ is odd, $c^{h-2}=\chi(c)c^{-2}$ for $c\in\mathbb{F}_q^*$. Therefore

$$\Pi_P=c_P^{h-2}\Big(-\frac{n_R}{n_Q}\Big)=\chi(F_Q(p_2)F_R(p_2))\,\frac{\Phi_R(P)}{\Phi_Q(P)},\qquad \Phi_W(U)=F_W(u_1)F_W(u_2),$$

where we used $\chi(c_P)=-\chi(F_Q(p_2)F_R(p_2))$. By symmetry,

$$\Pi_Q=\chi(F_R(q_2)F_P(q_2))\,\frac{\Phi_P(Q)}{\Phi_R(Q)},\qquad \Pi_R=\chi(F_P(r_2)F_Q(r_2))\,\frac{\Phi_Q(R)}{\Phi_P(R)}.$$

Now $\Phi_W(U)=\kappa_W^2\prod_{i,j}(u_i-w_j)$, so $\Phi_W(U)/\Phi_U(W)=\kappa_W^2/\kappa_U^2$, and the product of the three ratios of Φ's equals 1. The factors κ inside χ occur in pairs. Writing $a=p_2$, $a'=p_1$, $b=q_2$, $b'=q_1$, $c=r_2$, $c'=r_1$, we obtain

$$\Pi_P\Pi_Q\Pi_R=\chi\big[(a-b)(a-b')(a-c)(a-c')\cdot(b-c)(b-c')(b-a)(b-a')\cdot(c-a)(c-a')(c-b)(c-b')\big].$$

The three products $(a-b)(b-a)$, $(a-c)(c-a)$, $(b-c)(c-b)$ are negatives of squares and contribute $\chi(-1)^3=-1$. For the remaining six factors, compatibility of Q and R gives $\chi((b-c')(c-b'))=\chi((b-c)(b'-c'))$, and hence

$$\begin{aligned}&\chi\big[(a-b')(a-c')(b-c')(b-a')(c-a')(c-b')\big]\\&\quad=\chi\big[(a-b')(b'-c')(c'-a)\big]\,\chi\big[(a'-b)(b-c)(c-a')\big]=O(a,b',c')\,O(a',b,c),\end{aligned}$$

which is $\varepsilon(P,Q,R)$ by Lemma 4.1(ii). □

Corollary 4.3. *Let $q\equiv 3 \pmod 4$, $q\geq 7$, and let S be a complete exterior set which is an arc. Then $\varepsilon(P,Q,R)=-1$ for all distinct $P,Q,R\in S$. In the normal form of Section 3 this means*

$$\text{(A)}\qquad \chi(\psi(z)-\psi(y))=-\chi(z-y)\,\chi(yz)\qquad \textit{for all } y,z\in\mathbb{F}_q^*\ \textit{with}\ z\notin\{y,\psi y\}.$$

Proof. Let $S'=S\setminus\{P,Q,R\}$; then $|S'|=h-2$. For $X\in S'$ the line PX is a passant different from PQ,PR, and since S is an arc the map $X\mapsto PX$ is injective, hence a bijection onto the $h-2$ passants through P other than PQ,PR. Thus $\prod_{X\in S'}\lambda_P(PX)=\Pi_P$, and similarly for Q and R. No $X\in S'$ lies on a side of the triangle PQR, so multiplying (4.1) over $X\in S'$ gives $\Pi_P\Pi_Q\Pi_R=1$, and Lemma 4.2 gives $\varepsilon=-1$. The second statement follows from Lemma 4.1(iii) with $P=P_0$. □

5. Autocorrelations and Jacobi sums

In this section $\psi\colon \mathbb{F}_q^* \to \mathbb{F}_q^*$ satisfies (H1), (H2) and (A); $q \equiv 3 \pmod 4$ is an arbitrary prime power.

Lemma 5.1. *For all $x \in \mathbb{F}_q^*$ and all $z \in \mathbb{F}_q^* \setminus \{x, \psi x\}$:*

$$\chi\big(x(x-\psi x)\big) = -1, \tag{B}$$

$$\chi\big((x-\psi x)(z-\psi x)\big) = -\chi\big((x-\psi z)(z-\psi z)\big). \tag{C}$$

Proof. For (B), substitute $c = \psi(y)$, $y \in \mathbb{F}_q^*$, into $\sum_{c\in\mathbb{F}_q} \chi(c(\psi x - c)) = 1$, an instance of (W0) (the term $c = 0$ vanishes). For $y \notin \{x, \psi x\}$, (H2) and (A) give $\chi(\psi y)\chi(\psi x - \psi y) = \chi(x)\chi(x-y)$; this also holds for $y = x$ (both sides vanish), while for $y = \psi x$ the term equals $\chi(x)\chi(\psi x - x) = -\chi(x)\chi(x-\psi x)$. Hence

$$1 = \chi(x)\Big[\sum_{y\in\mathbb{F}_q^*} \chi(x-y) - 2\chi(x-\psi x)\Big] = \chi(x)\big[-\chi(x) - 2\chi(x-\psi x)\big] = -1 - 2\chi\big(x(x-\psi x)\big).$$

For (C), put $T_1 = \chi\big((x-\psi x)(z-\psi x)\big)$, $T_2 = \chi\big((x-\psi z)(z-\psi z)\big)$ and substitute $c = \psi(y)$ into $\sum_{c\in\mathbb{F}_q} \chi\big((\psi x - c)(\psi z - c)\big) = -1$ (W0). The term $c = 0$ equals $\chi(\psi x\, \psi z) = \chi(xz)$. For $y \notin \{x, \psi x, z, \psi z\}$, two applications of (A) give the term $\chi(xz)\chi\big((x-y)(z-y)\big)$, and this formula also gives the correct value 0 for $y \in \{x, z\}$. For $y = \psi x$ and $y = \psi z$ the terms are $U_1 = \chi\big((x-\psi x)(x-\psi z)\big)$ and $U_2 = \chi\big((z-\psi x)(z-\psi z)\big)$, whereas the formula would give $\chi(xz)T_1$ and $\chi(xz)T_2$. Now $U_i = T_i\,\chi\big((z-\psi x)(x-\psi z)\big)$, and (A) applied to the elements x and ψz (which lie in different pairs) gives $\chi(\psi x - z) = -\chi(x-\psi z)\chi(x\psi z)$, whence $\chi\big((z-\psi x)(x-\psi z)\big) = \chi(x\psi z) = -\chi(xz)$ and $U_i = -\chi(xz)T_i$. Using $\sum_{y\in\mathbb{F}_q^*} \chi\big((x-y)(z-y)\big) = -1 - \chi(xz)$ we get

$$-1 = \chi(xz) + \chi(xz)\big(-1-\chi(xz)\big) - 2\chi(xz)(T_1+T_2) = -1 - 2\chi(xz)(T_1+T_2),$$

so $T_1 + T_2 = 0$. □

For $\eta, \eta' \in \widehat{\mathbb{F}_q^*}$ put

$$M(\eta', \eta) = \sum_{x\in\mathbb{F}_q^*} \overline{\eta'(x)}\,\eta(\psi x),$$

the coefficients of $\eta \circ \psi$ in the character basis: $\eta \circ \psi = \frac{1}{q-1}\sum_{\eta'} M(\eta', \eta)\,\eta'$.

Lemma 5.2. *For all $\eta, \eta' \in \widehat{\mathbb{F}_q^*} \setminus \{1, \chi\}$,*

$$\big(J(\chi,\eta) + \overline{J(\chi,\eta')}\big)\, M(\eta', \eta) = -2(q-1)\,\delta_{\eta\eta'}.$$

Proof. Consider the matrices $\mathsf{A}_{xz} = \chi(x-z)$ (the Jacobsthal matrix) and $\mathsf{N}_{xz} = \chi(x-\psi x)[z = \psi x]$ indexed by $\mathbb{F}_q^* \times \mathbb{F}_q^*$, and the Hermitian product $\langle f, g\rangle = \sum_{x\in\mathbb{F}_q^*} \overline{f(x)}g(x)$. We have $(\mathsf{NA}+\mathsf{AN})_{xz} = \chi(x-\psi x)\chi(\psi x - z) + \chi(x-\psi z)\chi(\psi z - z)$. For $z = x$ this is $2\chi(-1) = -2$; for $z = \psi x$ it is 0; otherwise it equals $-(T_1+T_2) = 0$ by (C). Hence $\mathsf{NA}+\mathsf{AN} = -2I$.

The substitution $z = xt$ gives $\mathsf{A}\eta = J(\chi,\eta)\,\chi\eta$ for $\eta \neq 1$, and $\mathsf{A}^{\mathsf{T}} = -\mathsf{A}$ because $\chi(-1) = -1$. By (B) and (H2), $\chi(x-\psi x)\chi(\psi x) = 1$ and $\chi(x-\psi x) = -\chi(x)$, so $\langle \eta', \mathsf{N}(\chi\eta)\rangle = M(\eta', \eta)$ and $\langle \chi\eta', \mathsf{N}\eta\rangle = -M(\eta', \eta)$. Therefore

$$\langle \eta', \mathsf{NA}\eta\rangle = J(\chi,\eta)\, M(\eta', \eta),$$

$$\langle \eta', \mathsf{AN}\eta\rangle = -\langle \mathsf{A}\eta', \mathsf{N}\eta\rangle = -\overline{J(\chi,\eta')}\,\langle \chi\eta', \mathsf{N}\eta\rangle = \overline{J(\chi,\eta')}\, M(\eta', \eta),$$

and the sum equals $\langle \eta', -2\eta\rangle = -2(q-1)\delta_{\eta\eta'}$. □

6. Coincidences of Jacobi sums

In this section $q = p^k \equiv 3 \pmod 4$ and $q > 3$, so that $h = (q-1)/2 > 1$ is odd and k is odd. For $y \in \mathbb{Z}/h\mathbb{Z}$ put

$$E(y) = \sum_{j=0}^{k-1} (-1)^{\langle p^j y\rangle_h}, \qquad A(y) = \sum_{j=0}^{k-1} \langle p^j y\rangle_h.$$

Both are invariant under $y \mapsto py$, since $p^k = 2h+1 \equiv 1 \pmod h$. As $p^j < h$ for $j < k$, we have $A(1) = \sum_{j<k} p^j = 2h/(p-1)$.

Lemma 6.1. *Let $\beta \in \mathbb{Z}/h\mathbb{Z}$ and $\epsilon \in \{\pm 1\}$. If $E(u\beta) = \epsilon E(u)$ for all $u \in (\mathbb{Z}/h\mathbb{Z})^*$, then $\beta = \epsilon p^j$ for some $0 \le j < k$.*

Proof. Step 0: reduction to $\epsilon = 1$. The residues $\langle p^j\rangle_h = p^j$ are odd and $\langle -p^j\rangle_h = h - p^j$ is even, so $E(1) = -k$ and $E(-1) = k$. If $\beta = 0$, the hypothesis with $u = \pm 1$ gives $k = -\epsilon k$ and $k = \epsilon k$, which is impossible; so $\beta \neq 0$. For $z \not\equiv 0$ we have $\langle -z\rangle_h = h - \langle z\rangle_h$, so $E(-y) = -E(y)$ for $y \neq 0$. Replacing β by $\epsilon\beta$ we may assume $\epsilon = 1$.

Step 1: $A(v\beta) = A(v)$ for all units v. For $z \in \mathbb{Z}/h\mathbb{Z}$ write $\langle z\rangle_h/h = \sum_{\ell\ge 1} d_\ell(z) 2^{-\ell}$ with $d_\ell(z) \in \{0,1\}$, the binary expansion not ending in an infinite string of ones. Since $\langle 2w\rangle_h = 2\langle w\rangle_h - h\, d_1(w)$ and h is odd, $\langle 2w\rangle_h \equiv d_1(w) \pmod 2$; as $d_1(2^{\ell-1}z) = d_\ell(z)$, we get $\langle 2^\ell z\rangle_h \equiv d_\ell(z) \pmod 2$ for $\ell \ge 1$. Hence

$$E(2^\ell y) = k - 2c_\ell(y), \qquad c_\ell(y) = \#\{j < k : \ d_\ell(p^j y) = 1\}.$$

Applying the hypothesis to $u = 2^\ell v$ ($\ell \ge 1$, v a unit; 2 is a unit as h is odd) gives $c_\ell(v\beta) = c_\ell(v)$ for all $\ell \ge 1$, and therefore $A(v\beta) = h\sum_{\ell\ge1} c_\ell(v\beta) 2^{-\ell} = A(v)$.

Step 2: $\beta \in \{p^a\} \cup \{\frac{p^a+p^b}{2} : a \neq b\}$. Let $y_j = \langle p^j\beta\rangle_h$ for $j \ge 0$; then $y_k = y_0$ and $y_{j+1} = py_j - n_j h$ with integers $0 \le n_j \le p-1$. Summing over $0 \le j < k$ and using Step 1 with $v = 1$ gives $\sum_j n_j = (p-1)A(\beta)/h = (p-1)A(1)/h = 2$. Iterating, $y_0 = y_k = p^k y_0 - h\sum_{j<k} n_j p^{k-1-j}$, i.e. $2h\, y_0 = hN$ with $N = \sum_j n_j p^{k-1-j}$. Since the n_j are non-negative with sum 2, $y_0 = N/2$ is of the stated form.

Step 3: the second case is impossible. The set of β satisfying the hypothesis (with $\epsilon = 1$) is closed under multiplication by powers of p (as $E(py) = E(y)$) and under products $\beta\beta'$ with β' a unit (as $E(u\beta\beta') = E(u\beta') = E(u)$). Suppose $\langle\beta\rangle_h = (p^a + p^b)/2$ with $a \neq b$ (so $k \ge 3$). Multiplying by p^{-a} we may assume $\beta \equiv (1+p^c)/2$ with $0 < c < k$. Since $\gcd(p^c+1, p^k-1) = 2$ for odd k, β is a unit, so β^2 also satisfies the hypothesis and $\langle\beta^2\rangle_h$ is of the form found in Step 2. Note $2c \neq k$.

If $2c < k$, then $\langle\beta^2\rangle_h = (1 + 2p^c + p^{2c})/4 < h$, and $1 + 2p^c + p^{2c} \in \{4p^a,\ 2p^a + 2p^b\}$. This is impossible by uniqueness of base-p expansions: the left-hand side has the three non-zero digits $1, 2, 1$, the right-hand side at most two.

If $2c > k$, then $\beta^2 \equiv R := (p^k + 2p^c + p^{2c-k})/4 \pmod h$, because the difference $(1 + 2p^c + p^{2c})/4 - R$ equals $h\,(p^{2c-k} - 1)/2$. Moreover $h/2 < R < h$, the upper bound because $2p^c + p^{2c-k} + 2 \le 2p^{k-1} + p^{k-2} + 2 < p^k$. On the other hand, every element of the set in Step 2 is smaller than $h/2 = (p^k-1)/4$, except 3^{k-1} when $p = 3$. Finally $R \neq 3^{k-1}$: otherwise $2\cdot 3^c + 3^{2c-k} = 3^{k-1}$, which is impossible for $c = k-1$ by size and for $c \le k-2$ by uniqueness of base-3 expansions. Hence $\langle\beta\rangle_h = p^a$. □

Let $\eta_1 = \omega^{-1}$, a generator of $\widehat{\mathbb{F}_q^*}$; since ω is injective, so is η_1.

Proposition 6.2. *Let $\eta' \in \widehat{\mathbb{F}_q^*} \setminus \{1, \chi\}$. If $J(\chi,\eta_1) + \overline{J(\chi,\eta')} = 0$, then $\eta' = \chi\eta_1^{p^j}$ for some $0 \le j < k$.*

Proof. Write $\eta' = \omega^{-b}$ with $b \in \mathbb{Z}/(q-1)\mathbb{Z}$, $b \notin \{0, h\}$; note $\chi = \omega^{-h}$. As $\overline{J(\chi,\eta')} = J(\chi,\overline{\eta'})$ and $\chi\eta_1, \chi\overline{\eta'} \neq 1$, the hypothesis reads $G(\chi)G(\eta_1)/G(\chi\eta_1) = -G(\chi)G(\overline{\eta'})/G(\chi\overline{\eta'})$, i.e.

$$G(\omega^{-1})\, G(\omega^{-(h-b)}) = -\, G(\omega^{-(1+h)})\, G(\omega^b).$$

Applying σ_t for $t \in (\mathbb{Z}/(q-1)\mathbb{Z})^*$ and taking $\mathfrak{P}$-adic valuations (Theorem 2.2) gives

$$s(t) + s(t(h-b)) = s(t(1+h)) + s(-tb) \qquad \text{for all } t \in (\mathbb{Z}/(q-1)\mathbb{Z})^*. \tag{6.1}$$

(All four arguments are non-zero modulo $q-1$.)

Digit sums as sums of residues. If $\langle x\rangle_{q-1} = \sum_{i<k} x_i p^i$, then $\langle p^j x\rangle_{q-1} = \sum_i x_i p^{(i+j) \bmod k}$, because $p^k \equiv 1 \pmod{q-1}$. Summing over j gives $\sum_{j<k}\langle p^j x\rangle_{q-1} = s(x)\,(q-1)/(p-1)$. Consequently, for $x, y \in \mathbb{Z}/(q-1)\mathbb{Z}$,

$$s(x) + s(y) - s(x+y) = (p-1)\cdot \#\{j < k : \ \langle p^j x\rangle_{q-1} + \langle p^j y\rangle_{q-1} \geq q-1\}.$$

Since p^j is odd, $\langle p^j h\rangle_{q-1} = h$, and with $y = h$ we get $s(x) + s(h) - s(x+h) = (p-1)\gamma(x)$, where $\gamma(x) = \#\{j < k : \langle p^j x\rangle_{q-1} \geq h\}$.

Reduction to Lemma 6.1. For odd t we have $th \equiv h$, so $t(1+h) = t+h$ and $t(h-b) = h+(-tb)$. Substituting $s(t+h) = s(t)+s(h)-(p-1)\gamma(t)$ and $s(h+(-tb)) = s(h)+s(-tb)-(p-1)\gamma(-tb)$ into (6.1) gives $\gamma(t) = \gamma(-tb)$. The map $z \mapsto z+h$ on $\mathbb{Z}/(q-1)\mathbb{Z}$ exchanges $\{z : \langle z\rangle_{q-1} < h\}$ and $\{z : \langle z\rangle_{q-1} \geq h\}$, so $\gamma(x+h) = k - \gamma(x)$ for all x; and $z \mapsto -z$ does the same on residues different from $0, h$, so $\gamma(-x) = k - \gamma(x)$ whenever $p^j x \notin \{0, h\}$ for all j, i.e. $x \notin \{0, h\}$. Since $tb \notin \{0, h\}$, the condition becomes $\gamma(t) + \gamma(tb) = k$. For odd x and $z = \langle p^j x\rangle_{q-1}$ (odd), $z \geq h$ holds if and only if $\langle z\rangle_h = z - h$ is even; hence $\gamma(x) = \frac{1}{2}(k + E(x \bmod h))$ for odd x. Put $u = t \bmod h$ and $\beta = b \bmod h$. If b is odd, tb is odd and the condition reads $E(u\beta) = -E(u)$. If b is even, $tb + h$ is odd and $\gamma(tb) = k - \gamma(tb+h) = \frac{1}{2}(k - E(u\beta))$, so the condition reads $E(u\beta) = E(u)$. Every $u \in (\mathbb{Z}/h\mathbb{Z})^*$ lifts to an odd unit t, hence

$$E(u\beta) = (-1)^b E(u) \qquad \text{for all } u \in (\mathbb{Z}/h\mathbb{Z})^*.$$

Conclusion. By Lemma 6.1, $\beta = (-1)^b p^j$ for some $j < k$. If b is even, then $b \equiv p^j \pmod h$ and b even force $b = p^j + h = p^j(1+h)$ in $\mathbb{Z}/(q-1)\mathbb{Z}$, i.e. $\eta' = \omega^{-p^j(1+h)} = \chi\eta_1^{p^j}$. If b is odd, then $b = -p^j$ and $\eta' = \overline{\eta_1}^{p^j}$; but then $\overline{J(\chi,\eta')} = J(\chi, \eta_1^{p^j}) = J(\chi,\eta_1)$ by Lemma 2.1, and the hypothesis gives $2J(\chi,\eta_1) = 0$, contradicting $|J(\chi,\eta_1)| = \sqrt{q}$. □

7. The case $q \equiv 3 \pmod 4$

Throughout this section $q = p^k \equiv 3 \pmod 4$, $q \geq 7$, and ψ satisfies (H1)–(H3) and (A).

Lemma 7.1. *If $k \geq 3$, the integers $p^j - p^{j'}$ with $0 \leq j \neq j' < k$ are non-zero and pairwise incongruent modulo h.*

Proof. They are non-zero modulo h because $0 < |p^j - p^{j'}| \leq p^{k-1} - 1 < h$. They are pairwise distinct as integers: $p^j - p^{j'} = p^i - p^{i'}$ means $p^j + p^{i'} = p^i + p^{j'}$, and uniqueness of base-p expansions (digits at most $2 < p$) forces $(j, j') = (i, i')$. If $p \geq 5$, all of them have absolute value at most $p^{k-1} - 1 < h/2$, so distinct ones are incongruent. Let $p = 3$ and suppose $D_1 = 3^a - 3^b$ and $D_2 = 3^c - 3^d$ are distinct and congruent modulo h. As $|D_1 - D_2| \leq 2(3^{k-1} - 1) < 2h$, we get $D_1 - D_2 = \pm h$; after exchanging D_1, D_2 we may take the sign $+$, i.e. $2(3^a + 3^d) + 1 = 3^k + 2(3^b + 3^c)$. The right-hand side is at least $3^k + 4$; if $\min(a, d) \leq k-2$ the left-hand side is at most $2\cdot 3^{k-1} + 2\cdot 3^{k-2} + 1 < 3^k$. So $a = d = k-1$, and then $3^{k-1} + 1 = 2(3^b + 3^c)$. Reducing modulo 3 (recall $k \geq 3$) forces $b = c = 0$, hence $3^{k-1} = 3$, a contradiction. □

Proposition 7.2. *There are $c \in \mathcal{N}$ and $0 \leq j_0 < k$ such that $\psi(x) = c\,x^{p^{j_0}}$ for all $x \in \mathcal{S}$.*

Proof. By Lemma 5.2 and Proposition 6.2, $M(\eta', \eta_1) = 0$ for every $\eta' \notin \{1, \chi, \eta_1\}$ which is not of the form $\chi\eta_1^{p^j}$. Moreover $M(1,\eta_1) = \sum_x \eta_1(\psi x) = 0$ because ψ permutes $\mathbb{F}_q^*$, and $M(\chi,\eta_1) = \sum_y \chi(\psi y)\eta_1(y) = -\sum_y \chi(y)\eta_1(y) = 0$ by (H1), (H2). By Fourier inversion,

$$\eta_1(\psi x) = \alpha\,\eta_1(x) + \sum_{j<k} \beta_j\,\chi(x)\eta_1(x)^{p^j} \qquad (x \in \mathbb{F}_q^*)$$

for some $\alpha, \beta_j \in \mathbb{C}$, so on $\mathcal{S}$ we have $\eta_1(\psi x) = \sum_{j<k} \gamma_j\, \lambda_j(x)$, where λ_j is the restriction of $\eta_1^{p^j}$ to the cyclic group $\mathcal{S}$ of order h, $\gamma_0 = \alpha + \beta_0$ and $\gamma_j = \beta_j$ for $j \geq 1$. A character η_1^m is trivial on $\mathcal{S}$ if and only if $h \mid m$. By Lemma 7.1 (vacuous for $k = 1$) the characters $\lambda_j\overline{\lambda_{j'}}$, $j \neq j'$, of $\mathcal{S}$ are non-trivial and pairwise distinct. Since $|\eta_1(\psi x)| = 1$,

$$1 = \Big|\sum_j \gamma_j \lambda_j(x)\Big|^2 = \sum_j |\gamma_j|^2 + \sum_{j \neq j'} \gamma_j\overline{\gamma_{j'}}\, \lambda_j\overline{\lambda_{j'}}(x) \qquad (x \in \mathcal{S}),$$

and by linear independence of characters $\gamma_j\overline{\gamma_{j'}} = 0$ for $j \neq j'$. So exactly one coefficient γ_{j_0} is non-zero, and $\eta_1(\psi(x)x^{-p^{j_0}}) = \gamma_{j_0}$ for $x \in \mathcal{S}$. As η_1 is injective, $\psi(x)x^{-p^{j_0}}$ is constant on $\mathcal{S}$, equal to $c = \psi(1)$, which lies in $\mathcal{N}$ by (H2). □

Proposition 7.3. *If $\psi(x) = cx$ for all $x \in \mathcal{S}$, then $q \in \{7, 11\}$, and $c \in \{3, 5\}$ for $q = 7$, $c \in \{2, 6\}$ for $q = 11$.*

Proof. Since ψ is an involution, $\psi(cx) = x$, so the pairs of S other than P_0 are $\{x, cx\}$, $x \in \mathcal{S}$. For distinct $x, y \in \mathcal{S}$ and $t = x/y \in \mathcal{S} \setminus \{1\}$,

$$(x - y)(x - cy)(cx - y)(cx - cy) = c\,(x - y)^2\, y^2\, (t - c)(ct - 1),$$

so by (H3) and $\chi(c) = -1$,

$$\chi\big((t - c)(ct - 1)\big) = 1 \qquad \text{for all } t \in \mathcal{S} \setminus \{1\}. \tag{7.1}$$

If $c = -1$, then $(t-c)(ct-1) = -(t+1)^2$ with $t+1 \neq 0$ (as $-1 \in \mathcal{N}$), contradicting (7.1); note $\mathcal{S} \setminus \{1\} \neq \emptyset$. So $c \neq \pm 1$, and $f(r) = (r^2 - c)(cr^2 - 1)$ has the four distinct roots $\pm\sqrt{c}, \pm 1/\sqrt{c}$ in $\mathbb{F}_{q^2}$; in particular f is not a constant times a square. Splitting $\sum_{r \in \mathbb{F}_q} \chi(f(r))$ according to $r = 0$, $r^2 = 1$ and $r^2 \in \mathcal{S} \setminus \{1\}$ and using (7.1),

$$\sum_{r \in \mathbb{F}_q} \chi(f(r)) = \chi(c) + 2\chi\big(-(1 - c)^2\big) + 2(h - 1) = -1 - 2 + q - 3 = q - 6.$$

By (W), $q - 6 \leq 3\sqrt{q}$, i.e. $q \leq 19$. The prime powers $q \equiv 3 \pmod 4$ with $7 \leq q \leq 19$ are $7, 11, 19$. Table 1 lists, for each non-square $c \neq -1$, either a square $t \neq 1$ violating (7.1) or "admissible". □

Table 1. Non-squares $c \neq -1$ for $q = 7, 11, 19$ and the condition (7.1).

q	admissible c	excluded c, with a square $t \neq 1$ such that $\chi\big((t-c)(ct-1)\big) = -1$
7	3, 5	—
11	2, 6	$(c, t) = (7, 3),\ (8, 3)$
19	—	$(c, t) = (2, 4),\ (3, 6),\ (8, 7),\ (10, 4),\ (12, 7),\ (13, 6),\ (14, 9),\ (15, 9)$

Proposition 7.4. *The case $j_0 \neq 0$ in Proposition 7.2 does not occur.*

Proof. Suppose $\psi(x) = cx^\rho$ on $\mathcal{S}$ with $\rho = p^{j_0}$, $0 < j_0 < k$; in particular $k \geq 3$. For $x \in \mathcal{S}$, (B) gives $-1 = \chi(x(x - cx^\rho)) = \chi(1 - cx^{\rho-1})$; with $x = y^2$ we get $\chi(1 - cy^m) = -1$ for all $y \in \mathbb{F}_q^*$, where $m = 2(\rho - 1)$. Let $e = \gcd(m, q - 1)$ and choose u with $e \equiv mu \pmod{q-1}$. Then $y^e = (y^u)^m$, so $\chi(1 - cy^e) = -1$ for all $y \in \mathbb{F}_q^*$ and

$$\sum_{y \in \mathbb{F}_q} \chi(1 - cy^e) = 1 - (q - 1) = 2 - q.$$

The polynomial $1 - cX^e = -c(X^e - c^{-1})$ is separable because $p \nmid e$, so (W) gives $q - 2 \leq (e-1)\sqrt{q}$. On the other hand, e divides $\gcd(2(p^{j_0} - 1), 2(p^k - 1)) = 2(p^g - 1)$ with $g = \gcd(j_0, k)$. Since g is a proper divisor of the odd number k, $k/g \geq 3$; so $P = p^g$ satisfies $P \geq 3$ and $P^3 \leq q$, and $e - 1 \leq 2P - 3$. As $(P^3 - 4) - (2P - 3)^2 = (P - 3)(P^2 - P + 9) + 14 > 0$,

$$(q - 2)^2 \leq (e - 1)^2 q \leq (2P - 3)^2 q < (P^3 - 4)\, q \leq (q - 4)\, q < (q - 2)^2,$$

a contradiction. □

Theorem 7.5. *Let $q \equiv 3 \pmod 4$, $q \geq 7$. If a complete exterior set S of a conic in $\mathrm{PG}(2,q)$ is an arc, then $q \in \{7, 11\}$, and in the normal form with $P_0 \in S$ the involution ψ is $\psi(x) = cx$ for $x \in \mathcal{S}$ (and $\psi(x) = x/c$ for $x \in \mathcal{N}$), with $c \in \{3, 5\}$ for $q = 7$ and $c \in \{2, 6\}$ for $q = 11$.*

Proof. Move a point of S to P_0. By Corollary 4.3, ψ satisfies (A). Combine Propositions 7.2, 7.4 and 7.3. □

8. Proof of Theorem A

Part (a) is Proposition 3.2.

Part (b). By (a) we must decide when a complete exterior set S can be an arc. If $q \equiv 1 \pmod 4$, then S is linear by Theorem 1.1; as $(q+1)/2 \geq 3$, it is not an arc. If $q = 3$, S consists of two points, and the two exterior points on any of the three passants form such a set. If $q \equiv 3 \pmod 4$ and $q \geq 7$, Theorem 7.5 gives $q \in \{7, 11\}$; existence for these q is shown below.

Part (c) and existence. Let $q \in \{7, 11\}$ and $c_0 = 3$, respectively $c_0 = 2$. The pairs

$$q = 7: \ \{0,\infty\},\{1,3\},\{2,6\},\{4,5\}; \qquad q = 11: \ \{0,\infty\},\{1,2\},\{3,6\},\{4,8\},\{5,10\},\{9,7\}$$

are P_0 and the pairs $\{x, c_0x\}$, $x \in \mathcal{S}$. By the proof of Proposition 7.3 they form a complete exterior set (since c_0 satisfies (7.1), Table 1), and a direct computation of the $\binom{4}{3} = 4$, respectively $\binom{6}{3} = 20$, determinants of the corresponding forms (supplementary file `examples.py`) shows that it is an arc. The Möbius involution $x \mapsto 1/x$ fixes P_0 and maps $\{x, cx\}$ to $\{x', x'/c\}$ with $x' = 1/x \in \mathcal{S}$; so it conjugates the solution with constant c to the one with constant c^{-1}, and $3^{-1} = 5$ in $\mathbb{F}_7$, $2^{-1} = 6$ in $\mathbb{F}_{11}$. Hence, by Theorem 7.5, every arc-like complete exterior set is the image of the example above under an element of $\mathrm{PGL}(2,q)$, the group of collineations fixing $\mathcal{C}$ (for prime q this is the full stabiliser of $\mathcal{C}$). Computing the orbit of the example under $\mathrm{PGL}(2,q)$ (`examples.py`) gives orbits of length 14 and 22, i.e. stabilisers of orders $336/14 = 24$ and $1320/22 = 60$. Both stabilisers lie in $\mathrm{PSL}(2,q)$, and their element orders are distributed as $1, 9, 8, 6$ elements of orders $1, 2, 3, 4$ (for $q = 7$) and $1, 15, 20, 24$ elements of orders $1, 2, 3, 5$ (for $q = 11$); by Dickson's classification of subgroups of $\mathrm{PSL}(2,q)$ [9, Hauptsatz II.8.27] they are isomorphic to S_4 and A_5. Independently, an exhaustive search in $\mathrm{PG}(2,q)$ (`plane_check.py`) finds exactly 14, respectively 22, complete exterior sets that are arcs, and for $q = 7$ also confirms that these are the only $(q+1)/2$-sets S of exterior points for which $\mathcal{C} \cup S$ has no tangents. □

Remark 8.1. The stabiliser acts on the $(q+1)/2$ points of S as S_4 on 4 points ($q = 7$) and as $A_5 \cong \mathrm{PSL}(2,5)$ on 6 points ($q = 11$). The resulting sets without tangents of sizes 12 and 18 realise u_7 and u_{11} [16, 7].

9. Verification and concluding remarks

9.1. **Numerical checks.** All steps of the proof were checked numerically with short scripts (supplementary material; each runs in seconds): Proposition 3.2 by exhaustive search in $\mathrm{PG}(2,q)$, over all $(q+1)/2$-sets of exterior points for $q \leq 9$ and over all complete exterior sets for $q = 11, 13$; Lemma 4.2 on random triples for $q \leq 47$, including $q = 27$; Lemmas 5.1 and 5.2 on the examples for $q = 7, 11$; the combinatorial part of Proposition 6.2 (equivalence of (6.1) with $E(u\beta) = (-1)^b E(u)$, and the resulting set of b) exhaustively for $q = p \leq 107$ and for $q = 27, 243, 343, 1331, 2187$; Lemma 2.1 over $\mathbb{F}_{27}$ and $\mathbb{F}_{343}$; Proposition 7.4 directly for $q = 27, 243, 343$. Moreover, an exhaustive search (the C program `bsw_search.c` in the supplementary material, whose pruning rule is itself verified in Lean) lists all complete exterior sets for $q = 7, 11, 19, 23, 27, 31$; none of them is an arc for $q = 19, 23, 27, 31$, in agreement with the computations of Blokhuis, Seress and Wilbrink reported in [16, §3].

9.2. **Formal verification.** The main chain of the argument has been formalised in Lean 4 [5] with Mathlib [15] (Lean v4.35.0, Mathlib revision `56a0058`); see the supplementary material. The theorem `BSW.Main.main_vdv` states that for every finite field F of odd order $q \notin \{3, 7, 11\}$, no fixed-point-free involution of $\mathrm{PG}(1, q)$ with pairwise compatible 2-cycles (that is, no complete exterior set, Lemma 3.3) is an arc, where compatibility and collinearity are expressed by the formulas of Lemma 3.1. It is proved without `sorry` and without axioms beyond the standard ones (`propext`, `Classical.choice`, `Quot.sound`), but under explicit hypotheses expressing the classical results that are not in Mathlib:

- for $q \equiv 1 \pmod 4$: McConnel's theorem [11] (which contains Carlitz's theorem [4]), from which Theorem 1.1 is derived formally;
- for $q \equiv 3 \pmod 4$: the conclusion of Proposition 6.2, stated for an arbitrary generator η_1 of $\widehat{\mathbb{F}_q^*}$ (which is equivalent to the version above by Galois conjugation) and, for $k = 1$, in a slightly stronger form with any $\eta \notin \{1, \chi\}$ in place of η_1; and, for $k \geq 3$, Weil's bound (W) for square-free polynomials.

The geometric dictionary (Lemma 3.1(v)), Corollary 4.3 (by a different route, via determinants and Ceva's theorem), Lemmas 5.1, 5.2, 6.1, 7.1 and Propositions 7.2–7.4 are formalised; for prime q the formal proof replaces the use of (W) in Proposition 7.3 by an elementary argument. The examples for $q = 7, 11$ are checked by the Lean kernel. Not formalised are Stickelberger's theorem itself and the derivation of Proposition 6.2 from it, Proposition 3.2, and part (c) of Theorem A.

9.3. **Remarks.** Theorem A settles the case of arcs in the Blokhuis–Seress–Wilbrink conjecture: for $q \equiv 3 \pmod 4$, $q > 11$, no complete exterior set is an arc, whereas the conjecture predicts that every complete exterior set is linear for $q > 31$. The arc property enters our proof only through the identity (A), obtained from the Segre-type Lemma 4.2; for general complete exterior sets the sign ε is not constant, and the conjecture remains open.

Declarations

Data and code availability. The supplementary material — Python scripts for all computations mentioned in the paper, the search program and its output, and the Lean 4 project with its build and audit scripts — is available at `https://github.com/H1z3-br0/van-de-voorde-conjecture` and archived at Zenodo, doi:10.5281/zenodo.23670082.

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